

The Quest for a Resilient Internet Access in a Constrained Geopolitical Environment

Nowmay Opalinski
University Paris 8
Paris, France

Zartash Afzal Uzmi
LUMS University
Lahore, Punjab, Pakistan

Frédéric Douzet
University Paris 8
Paris, France

ABSTRACT

Six major submarine cable faults have severely disrupted business operations and public services in Pakistan in 2022. This is the largest number of network outages in a single year in the telecom history of the country, which routinely faces such blackouts. Using the example of Pakistan's Internet infrastructure, which is a typical example of a network overtly dependent on international bandwidth, we examine the relationship between resilient network services and various constraints imposed by technology, economics, and geopolitics. Through a combination of network mapping based on aggregated BGP data and face-to-face interviews with a variety of stakeholders, we identify key vulnerabilities in Pakistan's connectivity architecture and uncover the rationale behind connectivity decisions made by local Internet service providers (ISPs). We find that the lack of resilience of Pakistan's Internet is the result of power relations and friction between ISPs, international content providers (content delivery platforms), and state institutions. We also emphasize the geopolitical underpinnings of Pakistan's Internet connectivity and resilience.

CCS CONCEPTS

• **Social and professional topics** → **Import / export controls; Geographic characteristics.**

KEYWORDS

Resilience, Network Geography, International Connectivity

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1 INTRODUCTION

Availability of Internet connectivity has taken a central role in supporting all kinds of services our day-to-day lives now rely upon. From banking to ride-hailing, managing vacations to knowledge acquisition, booking healthcare appointments to social interactions, everything is fully dependent on one being “connected”. Any disruption in the network connectivity, particularly at a national scale,

directly impacts a large population. Physical disruptions of the network infrastructure may be bypassed with an alternate *physical* path that provides *resilience*. Such a path may not always exist. Many countries have what is typically known as *stub* type of international connectivity and a physical disruption may leave the entire country disconnected from the rest of the Internet [25].

This is what Pakistan experienced in 2022 when submarine optical fiber cuts left the country almost blacked out from the rest of the Internet, six times in that single year, severely downgrading the experience of local users. From 2002 to 2022, 24 recorded submarine cable cuts had a disruptive impact on the local Internet connectivity in the country.¹ In August 2017 after a cable cut occurred on the IMEWE submarine cable at a landing point in Jeddah,² Internet connectivity was downgraded for two days in Pakistan. It had an almost direct impact on the business operations of the national airline carrier, Pakistan International Airlines (PIA), and resulted in countrywide domestic and international flight delays [50]. Other testimonies also highlight severe disruptions to the operations of local start-ups due to the unavailability of e-banking services and online payments [38]. The impacts of such widespread connectivity disruptions are far-reaching, and the reason lies in the lack of resilient international network access. In Pakistan, the recurrent problem of network disruption affecting over 100 million Internet users has been mocked by local press and social media [32], many a time via widespread memes [9].

This situation is not limited to Pakistan. In March 2024, a break in several submarine cables off the West African coast led to a week-long disruption of Internet access in 13 African countries [43]. In fact, high dependency on international traffic is a rather widespread issue in emerging countries, where the lack of hosting infrastructure and reluctance of content platforms to set foot in what they deem as challenging environments, cause an over-reliance on the few paths connecting them to the global Internet.

In this context of international connectivity disruptions and limited opportunities for local content hosting, using Pakistan as a study case, we wish to answer the following research questions:

- RQ1:** Why does Pakistan's Internet offer no resilience against submarine cable cuts?
- RQ2:** How does the connectivity of Pakistan to the global Internet impact its network resilience?
- RQ3:** In which sense is Internet connectivity the product of power struggles between ISPs, international content-delivery platforms, and state institutions?



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¹According to a compilation of data we made from public declarations from the Pakistan Telecommunication Authority, Pakistan Telecommunication Corporation Limited (PTCL) and Transworld Associates (TWA).

²Then adding to the previous damages which occurred two months before on other submarine cables SMW4 and TW1.

Our first hypothesis is that the physical and logical architecture of Pakistan’s international connectivity, combined with the over-reliance on foreign content platforms, undermines its overall resilience. Thus, the lack of resilience is not just because of a disruption in international connectivity but also because the content of public interest is primarily hosted at foreign locations with limited local hosting, resulting in most of the national traffic constantly crossing the country’s borders [40]. Our second hypothesis is that this connectivity and application architecture stems from business and geopolitical factors that constrain technical decisions.

We highlight that Pakistan’s international connectivity lacks path diversity as it is highly concentrated at both the physical and logical layers. This leads to severe network downgrades when disruptions occur at crucial nodes and links. In general, the connectivity choices of the ISPs in Pakistan are heavily constrained. *First*, the ISPs fully rely on the content platforms to serve their customers. Thus, they face business constraints and always find themselves at the losing end of the negotiation table with the content providers. *Second*, and more importantly, the power relationships with the neighboring countries drive the external connectivity infrastructure of the entire country. Pakistan’s physical Internet landscape is closely scrutinized by state institutions and therefore shaped by their threat perceptions. As a result, cross-border terrestrial connectivity is strictly driven by security agendas. In other words, in Pakistan, technical wisdom is often overshadowed by political and security outlooks at the expense of network resilience. All in all, ISPs find themselves in a tough spot where the security outlook of the state compounded with business regulations (and the associated financial limitations) constrain their options.

In this paper, we tackle the above two factors—business constraints and geopolitical relationships—in answering our research questions. We reckon that attracting content platforms to bring their Points of Presence (PoPs) to Pakistan would free up the ISPs from the constraints put forth by the content providers. Currently, this is not the case. Geopolitical constraints are even harder to address but improved relationships may yield favorable results in terms of increased resilience against physical disruptions.

2 BACKGROUND

Resilience is an essential design and operational characteristic of every network within the global Internet [44]. Formally, it is “the ability of the network to provide and maintain an acceptable level of service in the face of various faults and challenges to normal operations”. In [44], Sterbenz *et al.* specify the parameters to consider for delivering resilience and survivability in communication networks. To prevent countrywide Internet downgrade from happening, technical wisdom is clear: ISPs need to increase the *redundancy*—the replication of paths to another ISP—and the *diversity*—the number of “different” paths to different ISPs—of their networks [1, 44].

With Pakistan as a representative case for other emerging countries, we consider resilience as the ability of the *national* network to maintain access to *international* connectivity. This restrictive definition of resilience is well-suited for countries like Pakistan because the overall user experience revolves around access to international platforms and services. A functioning domestic network, for instance, with no access to international services would still

be regarded as non-functioning from a user’s perspective. Therefore, in this paper, we discuss internal connectivity only marginally, leaving it as a topic of a separate study.

From a technology standpoint, independent administrative units, called the *autonomous systems* (ASes), interconnect via the Border Gateway Protocol (BGP), thus creating a global network—the Internet—where any user can connect to any other user worldwide. There may exist a direct physical connection between two ASes (A and B) without them having a logical connectivity (resulting from the lack of a BGP peering agreement), thus prohibiting a direct flow of data. The data may still traverse between A and B through another AS. Conversely if direct physical connectivity between two ASes is impacted due to a cut, but there exists a logical route between them as a result of a peering agreement, data packets must find an alternate physical route, possibly through other ASes.

The interconnection policies between ASes are normally governed by business interests but a wide range of non-technical parameters, driven by geopolitical relations, tremendously impact the connectivity choices of the ISPs which constitute the socio-technical object we call “the Internet” [28, 29]. In particular, these ISPs not only conform to the terrain characteristics for laying their infrastructure, they also adjust their commercial choices to the specific political economy they operate in, including the regulations imposed by state authorities. The ISPs are further constrained by the diplomatic relations of the state.

Recent literature [10, 24, 33, 36] on the geopolitics of digital networks and the larger reflections on digital sovereignty in political geography have clearly exposed how data routing architecture is impacted by governmental strategies. For example, the control over the network architectures of topological connectivity in China and Iran has enabled both governments to apply swift state censorship on their local informational sphere without having to disconnect from the entire Internet [36]. Such control is exercised through the concentration of connections at a few international gateways, enabling state authorities to easily apply filtering at the entry and exit points of their networks.

The control of the state over their network topology varies to different degrees depending also on their approach to digital sovereignty, which has emerged as a strong slogan for governments rejecting the dominance of US-based content platforms over the Internet and the absence of control over their citizens’ data [14]. Trends in data localization have been observed as local governments attempt to re-scale their control within cyberspace by bringing back territorial boundaries in the digital realm [23].

South Asia is particularly subjected to the rise of territorial borders in cyberspace because of persistent inter-state conflicts [49]. Most of these conflicts are caused by contested border delimitation following the partition of the subcontinent at the end of British rule [5]. This directly impacts the geography of the mobility of people and goods in the subcontinent. Pakistan, like India, essentially perceives its immediate terrestrial borders as a source of geopolitical anxiety rather than areas with a potential for economic opportunities [23], a conception that has a dire impact on South Asia’s geography of regional Internet connectivity. Pakistani authorities tend to strictly envision interactions with their direct neighbors through the lens of national security [39], resulting in very limited cyber-connectivity options.

3 METHODOLOGY

To assess the resilience of the Pakistani network, we mapped both the layers of connectivity: physical and logical. This helped us identify the vulnerabilities at the intersection of these two dimensions. This spatial approach to resilience is based on a methodology developed by researchers at Geopolitics of the Datasphere (GEODE) that combines a geographical analysis of the physical characteristics of the Internet and a spatialized analysis of routing policies using BGP data to create a cartography [11]. This mapping methodology helped us understand how geopolitics affect network resilience and how power relations impact user experience.

3.1 Physical Layer Mapping

We mapped Pakistan's physical fiber optic network architecture and its international connections. Mapping a country's physical network architecture is sometimes a challenge as this information is usually kept in-house by the telecom operators. However, compared to other countries, Pakistan offers valuable information that is openly available online. To get an overview of Pakistan's physical connectivity, we relied on four different sources: TeleGeography's submarine cable map for the undersea cables [46], the ITU Connectivity Infrastructure map [48], and additional maps from Pakistan's two main long-haul operators: Pakistan Telecommunication Company Limited (PTCL) and Wateen for terrestrial cables [17]. The compilation of this data allowed us to create a map of Pakistan's main fiber optic backbone network, including its points of entry and exit at the submarine cable landing stations and cross-border connection points.

3.2 Logical Layer Mapping

We mapped the logical structure of Pakistan's entanglements with the Internet by observing the BGP updates (announcements and withdrawals) made by the ASes registered in Pakistan. To this end, we captured the BGP updates crossing different ASes via a BGP observation platform developed by GEODE [11]. This platform generates a full graph of AS relationships every minute by processing up to 30 BGP flows from publicly available BGP routing data, RouteViews, and the RIPE Routing Information Service (RIS), which aggregates BGP messages from BGP monitors at cooperating ASes. Around 89,000 nodes and 200,000 links are included in the snapshot, supplemented by the relevant information such as the name of each AS, the country in which it is registered, and the IP address prefixes it announces. We also analyzed the number of prefixes and ASes associated with each country year on year, using the data retrieved from the Potaroo blog [34].

We used our BGP observatory to create a complete graph that gives us a snapshot of all Internet routes. We then applied filters to these snapshots to select the ASes that are relevant to our case study. Specifically, we selected (i) all the ASes registered in Pakistan, and (ii) all ASes which have at least one peering agreement with any AS registered in Pakistan. This allowed us to build a comprehensive understanding of how Pakistan is connected to the global Internet (a pictorial representation is provided in Appendix C). We also took a closer look at Pakistan's relations with countries within the region. For this purpose, we selected the ASes of Pakistan, its immediate neighbors (India, Afghanistan, Iran, and China), and three other

countries (Qatar, United Arab Emirates (UAE), and Oman) in the region. We studied the logical connectivity of these select regional countries among themselves (depicted in Appendix D) and with ASes outside this region (in Appendix E).

In spatialized graphs, each node is an AS and each link is a BGP agreement (see Figure 1 or Appendix C). The nodes are colored according to the country where the AS is registered. We used the Force Atlas 2 visualization algorithm to better visualize the degree of interconnection between nodes [18]. This is a graph spatialization algorithm that brings nodes with a high number of connections closer together, while nodes with few or no connections are moved away from each other. We also assigned a different size to each node by using a betweenness centrality metric [13]. The larger the node, the higher the proportion of shortest paths it has with all the other nodes in the graph. While the betweenness centrality metric is known to have its limitations when analyzing large-scale networks, it provides meaningful and reliable results when identifying potential bottlenecks in a rather small network (few hundred nodes and up to about 10,000 links).

Spatialized graphs are useful in explaining resilience because they allow researchers to identify phenomena of network concentration, which are known to be used by governments to facilitate DNS filtering, but that also have implications for routing agility [19, 41]. It also allows the *centrality* of ASes to be assessed to determine which ASes are critical to a country's connectivity [12].

3.3 Qualitative Interviews

We chose key stakeholders in the Pakistani telecom ecosystem and conducted 92 semi-directed interviews between March 2022 and March 2023 in three main cities of Pakistan (Karachi, Lahore, and Islamabad). This qualitative research initially focused on the operators of the national network, namely the 6 main fixed-line network providers and the 4 mobile network providers. In addition, we also spoke to the relevant authorities from the Pakistani government, including the Ministry of Information and Telecommunications, the Pakistan Telecommunication Authority (PTA), and the Universal Service Fund (USF). The interviews most relevant to this work were with Pakistani IP engineers, working at the international gateways of the only two international connectivity providers (PTCL and TWA), responsible for the day-to-day operation of international traffic over the country's submarine cables in Karachi. We also met with the Directors of Operations and Strategy of both companies at their headquarters in Islamabad. These qualitative interviews helped uncover the motivations of Pakistan's major ISPs for their connectivity decisions, and the constraints they face. These qualitative interviews enabled us to complete and better interpret the maps and graphs of the physical and logical connectivity.

4 SPATIAL NETWORK ANALYSIS

4.1 Karachi: the Neuralgic Point of Pakistan's Connectivity

A careful study of the fiber optic cable location data reveals that, like many countries in the Middle East and South Asia, Pakistan's international connectivity is almost entirely dependent on submarine cables that run through the world's busiest maritime trade routes connecting the Mediterranean Sea to the Indian Ocean and

further connecting the Indian Ocean to the South China Sea and the Pacific Ocean (see Appendices A and B). This has resulted in cables being concentrated around the major chokepoints in Egypt and Singapore. Pakistani connectivity is highly dependent on connections to Europe and is therefore vulnerable to any disruption of the submarine cables between Karachi and Marseille, where Pakistani ISPs source their European connections. Unfortunately, this route is one of the most congested trade routes in the world, leading to frequent disruptions of submarine cables, which are usually located in Egyptian territory, the Arabian Gulf, or the Mediterranean Sea [6] (a pictorial representation is provided in Appendix B).

This international submarine connection arrives in Pakistan via a single point: Karachi. At the time of our research, Karachi was connected to 6 submarine cables operated by PTCL and TWA, the only two companies owning the landing stations located in the seaport city. The concentration of landing points for submarine cables in a single location creates great vulnerability to physical or cyber incidents, whether intentional or not. Indeed, Karachi is also Pakistan's largest metropolis and the country's main commercial center and therefore attracts a high volume of maritime traffic. As a result, there have been additional cuts within Pakistan's territorial waters due to ships anchoring at the submarine cables. Such a concentration also creates a security risk if a terrorist group or other hostile force sabotages the landing stations or the main backbones connecting Karachi to the rest of the country, thus leaving most of Pakistan without access to international connectivity.

To make matters worse, there are almost no alternative terrestrial paths that connect to the global Internet. According to a publicly available map from PTCL, Pakistan's largest fiber backbone provider, there is only one fiber optic link to Pakistan's eastern neighbor India, connecting Lahore to Amritsar [17], but without any logical connectivity. On its western front, Pakistan is connected to Afghanistan at 3 different locations in Torkham, Bannu (Khyber Pakhtunkhwa), and Taftan (Balochistan), but Afghanistan itself is a land-locked country with even fewer connectivity options. The state of connectivity with Iran is difficult to assess based on publicly available maps. According to PTCL's main fiber optic infrastructure map, three connectivity points with Iran across the western border in Balochistan were in the pipeline, but there is no information on their completion. According to press reports, both governments discussed a potential cross-border connection in 2004 [7]. A new connection with China across the Khunjerab border was also established in 2017, as part of the China-Pakistan Economic Corridor (CPEC), which was presented as a new backup link for Pakistan in case of submarine cable disruption. Our exploration of Pakistan's physical network layer shows a strong concentration in Karachi. The logical layer analysis further endorses this observation.

4.2 Bottlenecks at the International Gateways

Figure 1 shows the logical routes between Pakistan and the rest of the global Internet as of January 1, 2023. To construct this graph, we selected the ASes registered in Pakistan and their immediate neighbors, eventually resulting in a final picture of 441 ASes, with 223 ASes registered in Pakistan, and 218 ASes registered abroad from 43 different countries. This graph shows that the Pakistani network is heavily concentrated on two nodes—its *points of control* [35]. The

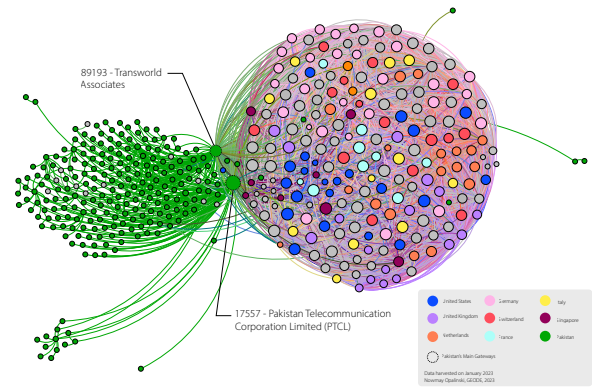


Figure 1: Pakistan's International Logical Connectivity. Complete BGP connectivity of all ASes registered in Pakistan. Each circle represents an AS, color-coded by country and sized by degree of connectivity. A magnified version is provided in Appendix C.

same two nodes (AS17557 for PTCL and AS89193 for TWA) play an important role in interconnecting Pakistan's ASes. We further note that these two ASes concentrate the majority of BGP agreements that are concluded with foreign ASes. They therefore play the role of gatekeepers for the country's international data flows. Almost all other ASes have to reach them to get to foreign destinations, they have a *de facto* duopoly over international transit. This concentration creates a vulnerability, as these two ASes are both the main internal hubs and the gateways to international connectivity. In other words, any physical or logical disruptions of PTCL's or TWA's ASes could lead to a nationwide network degradation.

Following this initial graph investigation, we also took a closer look at intra-regional connectivity by selecting ASes from Pakistan, India, Iran, Afghanistan, China, Qatar, Oman, and UAE to identify the existing logical paths connecting Pakistan with neighboring countries (Appendix D). We observed that the connectivity within the region is low: Pakistani ISPs mainly have peering agreements with their counterparts in the Arabian Gulf, to a lesser extent with ASes registered in Afghanistan and China, and no logical connectivity whatsoever with India and Iran.

The preliminary analyses of physical and logical maps helped us disclose key characteristics that explain Pakistan's lack of resilience: a high degree of network concentration at the internal and external ends, a high degree of dependency in terms of connections to Europe, and very few available pathways within its neighborhood. The spatial analysis allowed us to identify the main vulnerabilities. Then, through interviews, we were able to uncover the economic and geopolitical underpinning of these technical choices.

5 TECH AND BUSINESS CONSTRAINTS

In this section, we analyze the commercial reasons behind the technical decisions. Pakistan's BGP policies are primarily characterized by over-reliance on international content (traffic), a side effect of the dominant position of content providers on the Internet.

5.1 Content is the King!

The current reality of Pakistan's Internet connectivity is that more than 70% of national traffic is carried over GAFAM-like platforms^{3, 4}. According to figures from Telegeography's 2021 annual report 64% of the total international capacity used is focused on content provided by the GAFAM [45].

The key players currently influencing the shape of the global Internet are the content providers, in particular Google, YouTube, Meta/Facebook, WhatsApp, and Netflix. With bulk of their traffic going to these platforms, ISPs have no choice but to adapt their operations to the commercial objectives and deployment strategies of content providers. Thus, local ISPs are almost fully dependent on the content platforms, leading to constant negotiations to deliver their content to their own customers in the fastest and most reliable way. However, these negotiations depend on the goodwill of the platforms and their commercial calculations, which may run counter to the priorities of local companies and authorities. This is what we have observed in Pakistan.

In the early 2000s, international ISPs had no choice but to interconnect with servers in the US to access many Internet resources such as YouTube videos, causing latency issues for users who were far away from US networks. This led to global network outages once the undersea cables to the US were cut. To solve this accessibility problem, Internet platforms have developed two types of tools that are they now installing all over the world: Content Delivery Networks (CDNs) and Points of Presence (PoPs).

CDNs were first introduced by Akamai. The company found a clever way—replication—to bring content from distant servers closer to the end user. Other content delivery platforms followed Akamai's model and distributed their contents for free via the networks of Internet service providers. However, this practice has its own limitations. Only passive content such as videos, photos or text, can be replicated and stored. Active content, which plays an increasingly important role in our current Internet usage, such as video conferencing, live streaming or online games, must rely on dynamic interactions with the platforms' physical servers. Therefore, platforms are extending the CDN experience to a much higher level by expanding their physical node infrastructure across the Internet to replicate part of their core network in other locations via the Points of Presence (PoPs). Obviously, these two types of infrastructures differ in scope and reach. CDNs are essentially digital resources that can be easily stored on an ISP network, while PoPs in contrast, are large-scale physical infrastructure nodes located in a few places around the globe.

Since the advent of CDNs and PoPs, ISPs have been trying to bring this type of infrastructure to their networks and host it there to provide a better experience to their customers (both in terms of latency and reliability). To do so, ISPs have to engage in negotiations with the platforms in which they don't have the upper hand. They consult monthly reports from network monitoring tools to help them identify the directions of their outbound traffic. If they identify a new network that is taking up a significant amount of

bandwidth (e.g., a large increase in traffic directed to Amazon content outside of Pakistan), they then enter into a negotiation process to get better access to its resources, often with limited success. If the platform in question is an emerging resource of interest to local users, the local ISP does not necessarily have a peering agreement with that other network, which means it must connect to it via a third party—a transit provider—which in turn means it must pay a fee to reach the resource it needs. So local ISPs have different priorities when negotiating with a platform. Their preferred option is to ask them to host a local PoP or CDN on their own servers, which ultimately means better latency performance and a better experience for their users and also helps them gain an advantage over their local competitors.⁵ If the platform refuses, the ISPs at least try to enter into a peering agreement to obtain a direct, free data connection with the respective platform.

Until 2017, there were not enough CDNs in Pakistan, resulting in most traffic leaving the country to reach content hosted abroad [40]. An important step towards better resilience came as negotiations on the installation of platforms' *passive nodes* in the country succeeded. As of February 2023, all major Pakistani ISPs confirmed hosting CDNs (but not PoPs) from Google, Facebook, Netflix, and Akamai. Google CDNs have been set up in 8 locations in Pakistan (Karachi, Bahawalpur, Multan, Faisalabad, Lahore, Sialkot, Islamabad and Peshawar).⁶ The local regulator endorsed that more traffic is kept within Pakistan with increasing CDN deployment.⁷ Nevertheless, Amazon and Microsoft's Azure still do not have CDNs in the country. Even for Google and Meta, 'active mode' services (such as email and cloud) that require constant interaction between the end user and the origin server still cause data traffic in and out of the country for Europe or Singapore. The "million-dollar question"—as the Chief Operation Officer of one of Pakistan's largest ISPs dubbed it⁸—remains: Why are content providers refusing to install their Points of Presence (PoPs) in Pakistan?

5.2 Attracting Platforms' Infrastructure to Pakistan: an Unreachable Goal?

The question of why platforms are reluctant to set up PoPs in Pakistan leads to awkward answers and visible frustration on the part of Pakistani interlocutors. Content providers usually dismiss this possibility by invoking their lack of business interest at the moment. This is quite surprising knowing that Pakistan is the fifth most populous country in the world and that Internet connectivity is expanding at a rapid pace. Other reasons could be the key to this problem. One of the managers of Pakistan's international gateways mentioned that their refusal could be related to "international politics and other issues".⁹ Indeed, the Pakistani government is known to have a rather tense relationship with international platforms, mainly because the country blocks them whenever they host content that is deemed "unlawful". From various interviews and analysis of media reports, we conclude that large content providers are generally not interested in setting up their PoPs in Pakistan

³Interviews with IP gateway engineers from PTCL and Transworld Associates, Karachi, February 2023.

⁴Similarweb's list of Pakistan's most visited websites of Pakistan shows that the local Internet landscape is largely dominated by US platforms, with only Tik Tok coming at the 9th rank, see: Top Websites Ranking In Pakistan In August 2023 | Similarweb.

⁵Interviews with IP gateway engineers from PTCL and Transworld Associates, Karachi, February 2023.

⁶See: <https://cloud.google.com/cdn/docs/locations>.

⁷Interview with a PTA official, Islamabad, February 2023.

⁸Interview with Cybernet's COO, Karachi, February 2023.

⁹Interview with engineers of PTCL IP Gateways, Karachi, February 2023.

for various reasons: 1) Lack of legal framework, 2) Commercial disputes, 3) Privacy issues (the state demands to compromise it), 4) Emotion-driven censorship 5) Policies related to storage, 6) Lack of business incentives created by the state. In an interview with a representative of a content provider (who requested anonymity), it was revealed that the provider might consider setting up a PoP in Pakistan if the government promises to outsource all hosting of their content to that provider.

Most importantly, the lack of available infrastructure remains a major obstacle for content platforms in setting up their PoPs. One of their key demands is access to what they call “carrier-neutral data centers” or “carrier-neutral Internet Exchanges (IXs)” with enough space to host such nodes (PoPs).¹⁰ In formulating such a requirement, they refuse to host a PoP within an ISP’s own data center to avoid unequal treatment of customers. However, the data center market in Pakistan is structured in such a way that currently, only the telecom operators have the major data centers in Karachi, Lahore, and Islamabad and no other company offers alternatives.

Ultimately, these PoPs in Pakistan would help keep traffic within national boundaries and reduce pressure on undersea cable infrastructure; but this is not currently in sight. ISPs are increasing their submarine cable capacity due to the growing demand for international capacity, especially due to a sharp increase in the use of videoconferencing, which requires constant interaction with distant PoPs. However, Pakistani ISPs have few alternative cable routes and are still largely dependent on Internet bottlenecks in Egypt and Singapore for their international traffic.

In this constrained context, another solution for Pakistan would be to strengthen alternative terrestrial connections within its immediate neighborhood. However, as we will see in the next section, this is a daunting task, and digital cross-border land connections are still largely looked down upon from a security perspective.

6 GEOPOLITICAL CONSTRAINTS

6.1 Cross-border Connectivity Remains Strongly Subjected to Security-Bound Appraisal

In Pakistan, cross-border land communication is strictly subjected to authorization by the Pakistan Telecommunication Authority (PTA) and security clearance by the local security agencies, which have given few authorizations in this sense [22]. In short, cross-border communications and flows, including digital data, continue to be evaluated from a national security perspective, which largely hinders the prospects of diversifying international connectivity over terrestrial links. Consequently, Pakistan has very few active terrestrial links with its immediate neighbors (Appendix A).

6.2 The Paradox of Connectivity with India: Pakistan’s Greatest Perceived Threat Could also be the Best Option to Bring Resilience

The absence of connectivity between Pakistan and India is understandable considering that the physical borders of both countries are sealed and no direct trade is possible. However, a physical cable

does exist. Attempts to diversify data routes through Indian territory were discussed in Islamabad in the early 2000s. In 2005, after the week-long nationwide outage following the disruption of SMW 3 and STM-1 in Karachi Bay, it was considered to enhance global connectivity through the addition of further submarine cable links SMW-4 and TW1 and the construction of an alternative terrestrial connection point with India [2]. Taking advantage of the brief period of relaxed tensions between the two countries, PTCL entered into discussions with its Indian counterparts Bharat Sanchar Nigam Limited (BSNL) and Videsh Sanchar Nigam Limited (VSNL) to connect Lahore to Amritsar via the Wagah border crossing and then Mumbai, with a keen interest in reducing transit costs and creating more options for international transit through India [8].

From a technical point of view, many connections could be made across the 3300km border between the two arch-enemies. In fact, this would bring a lot of additional connectivity and redundancy to Pakistan’s network. As a senior officer of PTCL said: “People need to know about this, geopolitics has a heavy impact on our operation. We should be given the right to establish more connectivity with India, it would help us to improve the overall resilience of our network”.¹¹ This makes technical sense as the majority of Pakistan’s population lives in the eastern part of the country along the Indus River and in Punjab, regions that are densely connected and very close to other Indian cities. These two factors suggest that connecting the two networks across the border would not be too costly for local operators. Such a link could further reduce the cost of entry into the international transit business for many Pakistani operators who do not have the resources to invest in a costly undersea link. But then geopolitics comes into play.

Though the terrestrial cable was commissioned in 2008 and was about to be operational starting from 2009—the connection was even tested and worked—no commercial data ever crossed the border through it due to the final opposition from security apparatus on both sides of the border [47]. According to our interviews with Pakistani ISPs, there is no law prohibiting traffic exchange with India, but there is an unwritten ban on direct data sharing with India or Israel. It is thus no surprise that Pakistani ISPs are unable to interconnect at Indian IXPs to peer with other content providers. These providers often negotiate with Pakistani ISPs and propose to peer with their PoPs in Mumbai to get lower latency.¹² In fact, India is emerging as one of the major hubs for Internet platforms and cloud providers in Asia, as US-based content providers have rushed to establish PoPs in the country and have even made India their rear base for cloud services provision in South Asia. These platforms also have multiple edge locations (PoPs) in the country, 6 for Amazon AWS, 6 for Microsoft Azure and 5 for Google.¹³

GAFAM’s network planning priorities in the region therefore appear to be at odds with the Pakistani authorities’ desire to promote local hosting and protect national communications from possible Indian interception. In fact, the Pakistani authorities do not consider the benefit of greater redundancy through interconnection at multiple points across the India-Pakistan border.

¹¹Interview with PTCL’s COO, Islamabad, February 2023.

¹²Interview with PTCL IP engineers, Karachi, February 2023.

¹³See Google’s cloud edge-location map: <https://cloud.google.com/vpc/docs/edge-locations?hl=fr>

¹⁰Interview with a PTA official, Islamabad, February 2023.

Pakistani authorities are also further securing their communications from India, by avoiding infrastructure that passes through India's territorial waters. They also plan to build a new landing station at the Chinese-built port of Gwadar, which would provide an additional link and less vulnerability to potential security threats from New Delhi. Early press reports revealed that the National Telecommunication Corporation (NTC), which is responsible for providing connectivity to government and defense establishments in Pakistan, is implementing the project, which is also part of the China-Pakistan Economic Corridor (CPEC) [30]. According to our interviews, this station should not be operational before 2026. Back in 2017, the CEO of PTCL mentioned that the company was aiming for a new submarine cable link with a landing station in Gwadar in 2018, which has still not happened [50].

Bangladesh, the former eastern wing of Pakistan before it became independent in 1971, is an interesting point of comparison for Pakistan's security-focused network planning. The country reacted in the opposite way after a nationwide Internet blackout in 2012 caused by a disruption of its only undersea cable at the time, Sea-Me-We 4, near Singapore. In the same year, the Bangladeshi government decided to establish 6 cross-border terrestrial connections with India as a backup option by issuing permits for terrestrial Internet cables. These cables have been in operation since then and have been used by Indian ISPs since February 2016 to establish international connections via Bangladesh and connect the poorly served northeastern provinces of India. Ultimately, Bangladesh has successfully created fallback options and even generated revenue by becoming a transit provider for its larger neighbor [21].

6.3 Afghanistan and Iran: Potential Clients or Alternative Connection Paths?

Pakistan has been and remains the most important source of international connectivity for Afghanistan, which also gets small amounts of bandwidth from its Central Asian neighbors and Iran. The main reason is that Pakistan, which shares a 2670 km border with Afghanistan, offers the shortest route for the landlocked country to the submarine cables, with several Pakistani ISPs selling bandwidth to main Afghan ISPs. The feasibility of routing through Pakistan is further supported by the fact that Afghanistan's main demand for connectivity comes from the regions bordering Pakistan. This has given Pakistan important leverage on Afghanistan's international transit, but the picture could change as Afghan ISPs are currently trying to diversify their routes through Central Asia and Russia.¹⁴ Although Pakistani ISPs sell bandwidth to their Afghan counterparts, meaning there is a direct logical path between the two countries' networks (Appendices D and E), traffic from Pakistan to Afghanistan does not find a logical link in that direction and is often routed through Central Asia, Russia, and Europe, showing a low degree of intra-regional connectivity [4].

On its southwest, Pakistan shares a 909 km border with Iran. Both countries, along with Turkey, were part of an extensive—now decommissioned—microwave phone network, sponsored by the USAID in the 1960s. However, it is not clear whether Pakistan exchanges direct Internet traffic with Iran. The ISPs of both countries have laid their fiber optic cables up to the border, but the two fiber

optic networks are still not physically connected. It means that such cross-border traffic could be open if the state authorizes it. According to press reports, both governments discussed potential cross-border connectivity in 2004 [7]. According to Pakistani operators, there are currently several reasons for the lack of cross-border data traffic with Tehran. First, Pakistani companies do not want to be confronted with extraterritorial US sanctions, as Iran is affected by international sanctions. Second, the supplier of fiber optics on the Pakistani side has objected to using its hardware to connect to Iran for the same reasons. Third, the allegedly high tariffs for transit from the Iranian side do not make economic sense for the Pakistani stakeholders. Moreover, according to our interviews, Pakistani players have no idea about the connectivity situation in Iran and how the country is connected to the global Internet. Therefore, they do not consider Iran as a viable option for a backup connection. Our interviews reveal that Pakistani ISPs perceive Iran in the same way they see Afghanistan, namely as a potential capacity buyer that could use Pakistan as a transit to submarine connections.

6.4 The Chinese Path: Viable Alternate Route?

Even though the Chinese border is the hardest to reach, Pakistan is investing a lot of resources and efforts in its cross-border connectivity with China along the Khunjerab Pass in the high mountains of the Karakoram. Geopolitics is paramount in this connectivity choice as well. Pakistan and China have been very close partners since the Sino-Indian War of 1962, with a shared perception of India's hostility towards their interests being the main driver of their partnership [42]. In the 1970s, the construction of the world's highest cross-border highway, the Karakoram Highway, linking Gilgit to Kashgar in Xinjiang, symbolically opened new avenues for state-sponsored interactions between the two countries [20]. A cross-border fiber optic cable between Pakistan and China was planned in 2007, during the military rule of General Pervez Musharraf, with the signing of a memorandum of understanding (MoU) at the state level. Due to financial problems and technical difficulties, it was not implemented until a decade later, in 2017. The launch of CPEC in 2015 provided an opportunity to include the cable under the big umbrella of the 37 billion dollar BRI flagship project along the China-led transportation infrastructure and power plants. The fiber-optic link was therefore included in the CPEC Long-Term Plan and financed by a loan from China's EXIM Bank.

Equipped with Huawei hardware, this new 820 km long cable connects the Pakistani Army's civil telecommunications wing, the Special Communications Organization's (SCO) Headquarter in Rawalpindi to the border with China, at Khunjerab pass at 4693m above sea level. The Pakistani section of the cable, owned by the SCO, was showcased as a new backup link for Pakistan, helping to mitigate the risk of a nationwide network downgrade by providing additional transit options through China [15, 16]. In a speech to a Senate committee tasked with assessing interest in the project, the then SCO director justified the project by citing the need to create a new link outside of Indian interference. He even stated that previous submarine cables were laid under consortia involving Indian companies and therefore jeopardized Pakistan's national security. He also said that "some incoming and outbound Internet traffic landed in India before being routed to its destinations,

¹⁴Interview with PTCL's COO, Islamabad, February 2023.

posing a security risk for Pakistan” [37]. This statement reiterates the importance of security-related considerations in the planning of cross-border network accessibility. According to press releases from newspapers that had access to the confidential CPEC Long-Term Plan, including the section on telecommunications, securing communications was the primary focus of the construction of the Pak-China OFC Cable [15, 16]. This securitization goal would serve both partners by providing a secure transit corridor for Chinese communications with African countries and the Middle East, as well as helping Pakistan reduce its dependence on submarine cables.

Pakistan is well-placed to provide alternate, possibly more secure, transit to China. For example, with the commissioning of the PEACE Cable, a new shortest route linking China to East Africa and Europe through Pakistan is emerging. Likewise, China Mobile has invested in another submarine cable, Africa1, which also gets its final Eastern landing station in Pakistan, and which will connect Africa along its coast from North to South and West to East. While this link clearly serves Chinese interests, it is still unclear if that additional link will provide a true backup alternative to Pakistani ISPs.

6.5 The Emerging Role of Arabian Gulf Countries as Regional Connectivity Hubs: An Opportunity

The best connectivity options for Pakistani operators are offered by the Gulf countries. According to our interviews, Pakistani operators prefer to connect with the UAE, Oman, and Saudi Arabia. The last major disturbance on submarine links has proven this to be the right choice. In November 2022 the simultaneous cut of 3 submarine cables in Egypt had resulted in a limited downgrade of Pakistan’s connectivity because ISPs first could alleviate the disruption by switching their traffic towards Singapore, even if it meant more latency, and by redirecting some services towards the Arabian Gulf.¹⁵ While it still lacks a wide panel of content peering options such as Western Europe and Singapore, the Arabian Gulf is progressively becoming a regional connectivity hub where international content platforms increasingly settle their PoPs.

The competition of Gulf countries over network resources from influential platforms is changing the connectivity landscape in the region. Countries that have until recently played a marginal role in international Internet transit are emerging as regional connectivity crossroads (see [10] for a detailed discussion). Such a strategy emerged early on in Bahrain, followed by UAE and Oman, where local elites progressively undertook a diversification strategy of their revenues outside of the traditional oil trade. The development of digital infrastructure sits at the core of their attempts to spur the IT services sector in their country [3, 26, 31]. Local and international stakeholders have invested tremendous sums in the building of large-scale data centers located at very advantageous locations, in the coastal cities of Fujairah (UAE) or Muscat (Oman) where submarine cable stations are already established.

A relatively latecomer to that game, Saudi Arabia, has also shown ambitions to become a major transit provider in the region. In July 2021 Google disclosed two new submarine cable projects linking Europe to the Middle East. Whereas the Blue cable will interconnect Europe with Israel, more importantly, the Rama cable will link

Western Europe to India by crossing and landing in Jordan, Saudi, and Oman territories. This new path could finally bring to an end Egypt Telecom’s very contested monopoly over the transit between Europe and Asia [27]. In 2022 the kingdom announced that its state-owned operator, Saudi Telecom Corporation (STC), will transfer its data centers, submarine cables, and Points of Presence (PoPs) assets to a specific wholly-owned entity in charge of developing that specific domain of activity.

For Pakistani ISPs, interconnecting to content platforms in the Gulf would ultimately reduce their dependency on Europe-bound traffic and more crucially the highly vulnerable path going through the Red Sea. These new dynamics highlight a form of regionalization of data traffic which could lead to an eventual de-concentration of digital transit fluxes from powerful switching crossroads sitting in Western Europe. From Pakistan’s perspective, despite Islamabad’s data localization policy failure, increased connectivity with the Arabian Gulf seems to be the best ultimate choice.

7 CONCLUSIONS

Submarine cable cuts disrupt Pakistan’s Internet access, with no resilience whatsoever, for two major reasons: 1) the dominance of content delivery platforms, so much so that they drive the technology/business decisions for local infrastructure, and 2) the decision of state actors to maintain restrained cross-border connectivity.

Our geographical exploration of Pakistan’s global Internet connectivity illustrates that the structure of such connectivity, and thus the underlying resilience, is driven by the relationship its operators maintain with content delivery networks and content platforms globally. As Pakistan suffered from major Internet blackouts, ISPs and local authorities implemented several strategies ranging from an increase in submarine cable diversity to hosting more content locally. However, important limitations remain because of the lack of interest from content platforms in hosting crucial active nodes within Pakistan. Our study thereby underlined the power asymmetry between platforms, operators, and regulators from a rather understudied perspective: how the infrastructure planning of content platforms indirectly shapes the geography of a country’s Internet connectivity structure.

Apart from this business and technical influence, geopolitical representations bear a significant footprint on the geography of cyberspace. Political proximity and strategic partnerships shape Pakistan’s cross-border Internet traffic, sometimes even against technical wisdom. Thus, a country’s outreach to the Internet is a continuous struggle and revolves around negotiated choices.

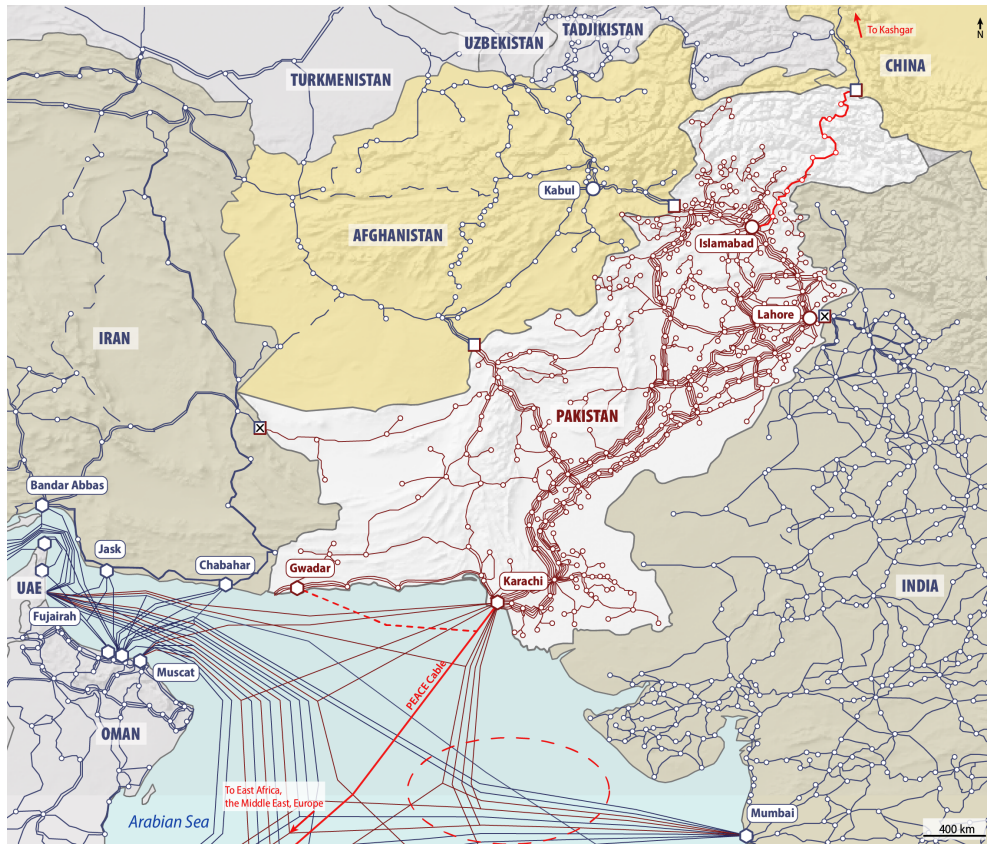
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¹⁵Interview PTCL and Transworld IP gateways engineers in Karachi, February 2023.

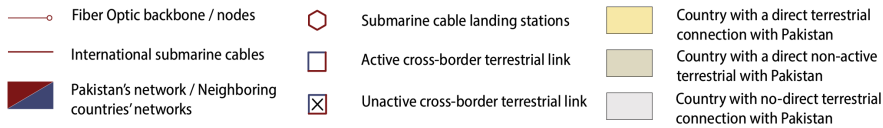
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A PAKISTAN'S PHYSICAL CONNECTIVITY WITH NEIGHBORS



**Pakistan's Internet network interconnections with its neighborhood:
a low degree of integration and emerging links with China along CPEC**

Pakistan's few active connections with its direct neighbors result in a high dependency on international submarine cables ...



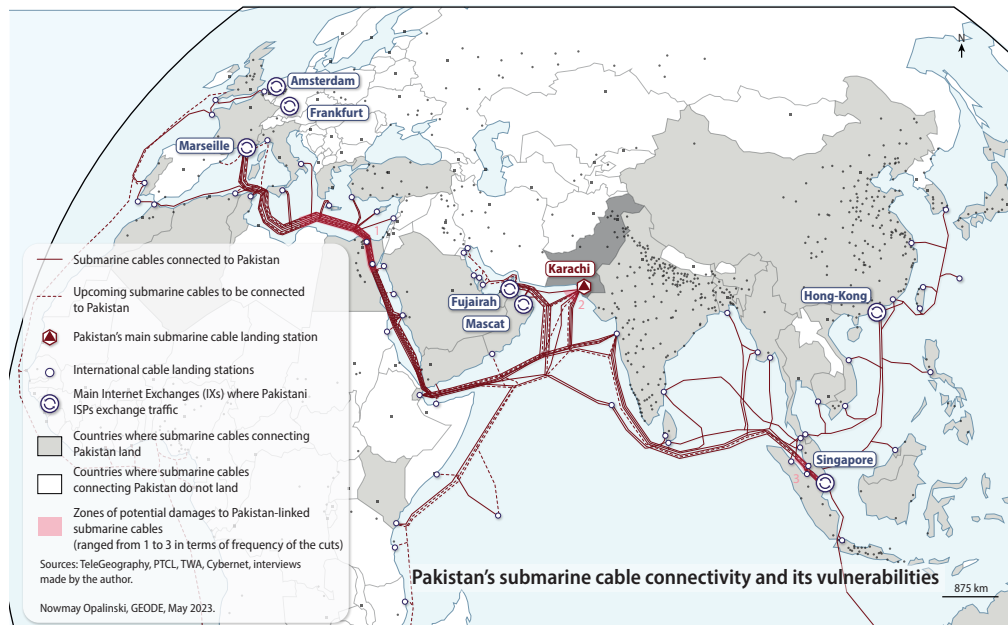
... new alternative China-sponsored paths should help bringing more resiliency and securing sensitive communications.



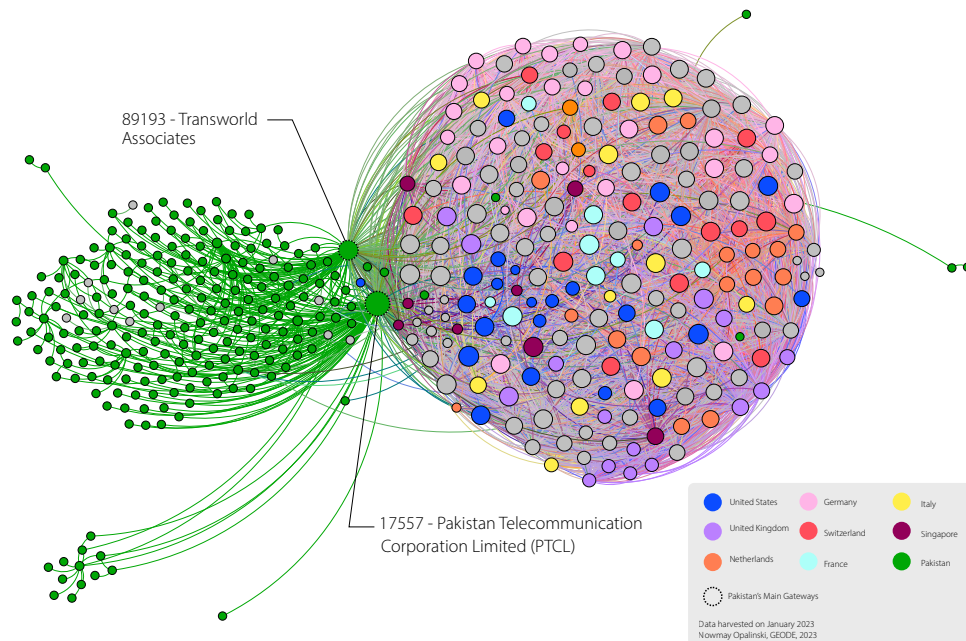
Sources : ITU, Telegeography, Pakistan Telecommunication Authority, PTCL
interviews with Telecom engineers in the field.

Conception and realization: Nowmay Opalinski (GEODE/IFG), June 2023.

B PAKISTAN'S SUBMARINE CABLE CONNECTIVITY AND ITS VULNERABILITIES

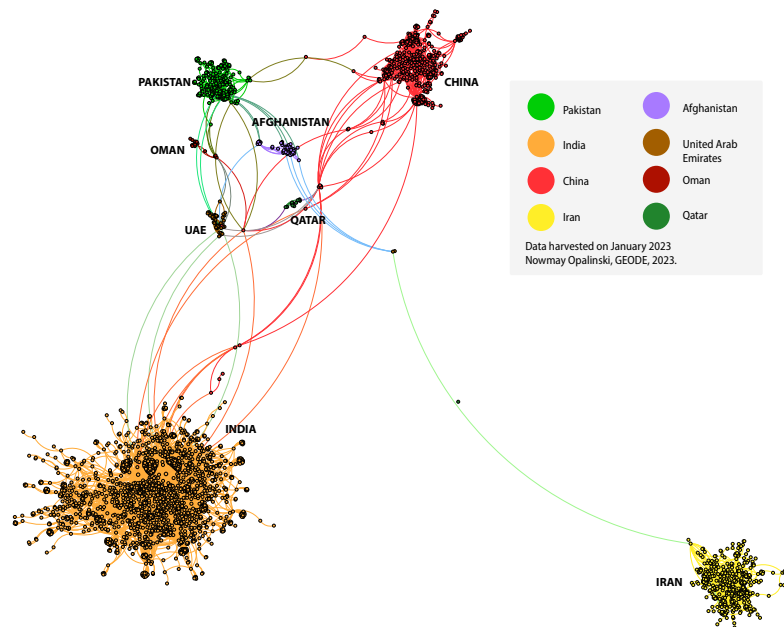


C PAKISTAN'S INTERNATIONAL LOGICAL CONNECTIVITY



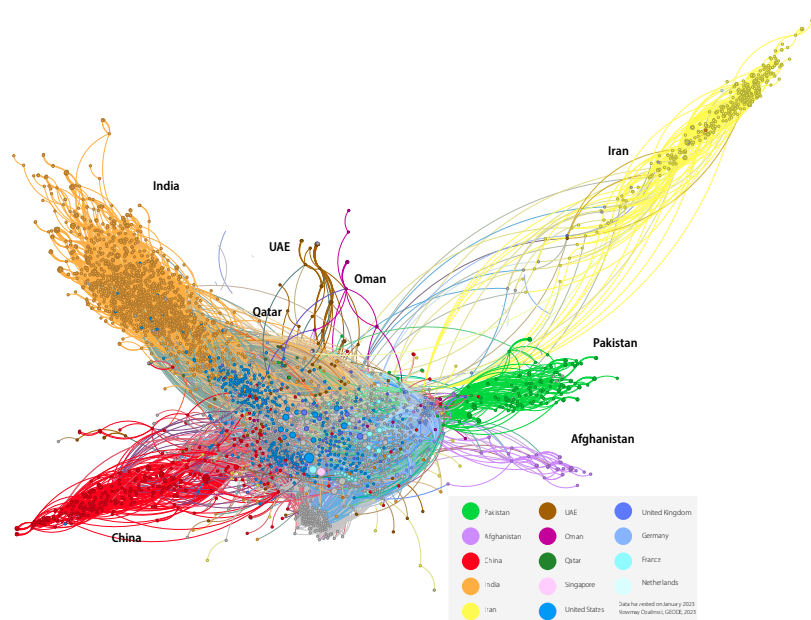
Complete BGP connectivity of all ASes registered in Pakistan. Each circle represents an AS, color-coded by country (green for Pakistan) and sized by betweenness centrality (degree of connectivity). The data is obtained from the BGP observation platform developed by GEODE and plotted with Gephi using the Force Atlas 2 visualization algorithm. Section 4.2 provides the details.

D INTRA-REGIONAL LOGICAL CONNECTIVITY



Illustrating the BGP agreements between the autonomous systems registered in some of the regional countries (India, Afghanistan, Iran, China, Oman, UAE and Qatar)

E FULL LOGICAL CONNECTIVITY OF REGIONAL PLAYERS



Illustrating all BGP agreements of the autonomous systems registered in regional countries (India, Afghanistan, Iran, China, Oman, UAE and Qatar). ASes registered in Europe play a key role in providing transit within the region.